Product Requirement Document

ShiftNet — A WordNet browser that shows how word meanings shift over time

1. Vision & Goals

💡 *Explain the problem, your solution, and who you are building this product for.*

Problem

WordNet browsers such as Open English WordNet (en-word.net) list a word's senses as a flat, timeless set. Nothing shows that the computing sense of *cloud* grew out of the weather sense, or when that happened. Researchers who study semantic change rebuild these histories by hand from dictionaries and corpora.

Solution

ShiftNet keeps the familiar en-word.net experience: search a word, then read its synsets, definitions, examples and relations. It adds time. Senses are listed by registration date, and a sense born from an earlier one branches off it, like commits in a git tree visualizer. Readers see at a glance how a word split, narrowed or drifted.

Target users

- Historical linguists and lexicographers studying semantic change
- Computational linguists and NLP researchers who need dated, diachronic sense data
- Teachers and students exploring etymology and language change
- Curious readers who want the story behind a word

2. User Stories

💡 *Each user story = 1 feature. Format: "As a X, I want Y so that Z". One story = one screen or one action.*

US1. **Home**
As a visitor, I want a home page with a search box and the latest registered senses so that I can start exploring right away.

US2. **Search**
As a user, I want suggestions while I type a word so that I reach its entry even with partial spelling.

US3. **Word entry**
As a user, I want to see all senses of a word grouped by part of speech, as on en-word.net, so that I know what the word means.

US4. **Sense lineage**
As a user, I want a word's senses listed by registration date, each linked to the sense it was born from like a git tree, so that I can see how its meaning shifted.

US5. **Global timeline**
As a user, I want to browse senses across the whole lexicon by registration date, in the same git-tree style, so that I can discover new meanings as they appear.

US6. **Sense detail**
As a user, I want to open a sense and see its definition, examples, date, parent sense, shift type and source so that I can check the evidence.

US7. **Relations**
As a user, I want to follow a sense's WordNet relations (synonyms, hypernyms, hyponyms and others) so that I can explore related words.

US8. **Time travel**
As a user, I want to pick a date so that I see a word's senses as they stood at that point in time.

US9. **Filters**

As a user, I want to filter senses by part of speech, shift type and date range so that I can focus on one pattern of change.

US10. **Share & cite**

As a user, I want a stable link and a ready-made citation for every word and sense so that I can reference them in my work.

3. Acceptance Criteria

💡 *The conditions for a user story to count as “done”. The AI uses them as a checklist.*

US1 — Home

- ☐ The search box is centred and focused when the page loads
- ☐ The 10 most recently registered senses are listed below it, each with word, short gloss and date
- ☐ A link opens the full global timeline

US2 — Search

- ☐ Suggestions appear after 2 characters
- ☐ Suggestions include multi-word expressions (e.g. “hot dog”) and show the part of speech
- ☐ Search ignores case and accents
- ☐ A word with no match shows “No results” plus close spellings

US3 — Word entry

- ☐ Senses are grouped by part of speech: noun, verb, adjective, adverb
- ☐ Each sense shows its synonyms, definition and examples when available
- ☐ Each sense shows its registration date
- ☐ A toggle switches between the Senses view and the Timeline view (US4)

US4 — Sense lineage

- ☐ One sense per row, sorted by registration date, oldest first (toggle: newest first)
- ☐ A sense born from an earlier sense starts a new branch from its parent’s node, as in `git log --graph`
- ☐ A sense with two parents is drawn as a merge of both branches
- ☐ Each branch has its own colour; a hollow node marks an obsolete sense and ends its branch
- ☐ Hovering or focusing a node highlights its full ancestry back to the root sense
- ☐ A parent sense that belongs to another word appears as a link to that word at the start of the branch

US5 — Global timeline

- ☐ Rows list registered senses across the lexicon, newest first, with year separators
- ☐ Same graph rules as US4; lines connect only senses of the same lineage
- ☐ A branch whose parent is outside the loaded rows shows a short “from: parent sense, date” stub instead of a long line
- ☐ Infinite scroll loads 50 rows at a time; a “Jump to year” control moves through time

US6 — Sense detail

- ☐ Shows lemma, part of speech, definition, examples, registration date, date source and shift type
- ☐ Parent and child senses are listed as links
- ☐ A mini lineage graph shows the sense within its word’s tree, with the current sense highlighted

US7 — Relations

- ☐ Shows the synset’s relations: synonyms, antonyms, hypernyms, hyponyms, meronyms, holonyms, derived forms
- ☐ Each related word links to its entry
- ☐ Relation groups with more than 10 items are collapsed by default

US8 — Time travel

- ☐ A date slider spans the word’s earliest to latest registration date
- ☐ Senses registered after the chosen date are hidden; senses obsolete by then are greyed out
- ☐ The chosen date is kept in the URL

US9 — Filters

- ☐ Filters: part of speech, shift type (broadening, narrowing, metaphor, metonymy, amelioration, pejoration, other) and date range
- ☐ Active filters show as removable chips
- ☐ Filtered-out senses keep their lines as faint dashed connectors, so the lineage stays readable

US10 — Share & cite

- ☐ Every word, sense and timeline state has a stable URL
- ☐ A “Cite” button copies a citation in APA and BibTeX formats
- ☐ Opening a shared URL restores the same view, filters and date

4. Application Screens

💡 *List every screen. This helps the AI understand the app’s structure and navigation.*

Screen	Description
Home	Search box + latest registered senses
Search results	Matching lemmas with part of speech and number of senses
Word entry — Senses	Senses grouped by part of speech, en-word.net style
Word entry — Timeline	Senses by registration date with the git-tree lineage graph
Sense detail	Definition, examples, date, source, parents, children, relations
Global timeline	All senses across the lexicon by date, git-tree style
About & sources	Project description, data sources, how dates are assigned, how to cite
Not found	Unknown word or sense, with spelling suggestions

Key screen: Word entry — Timeline

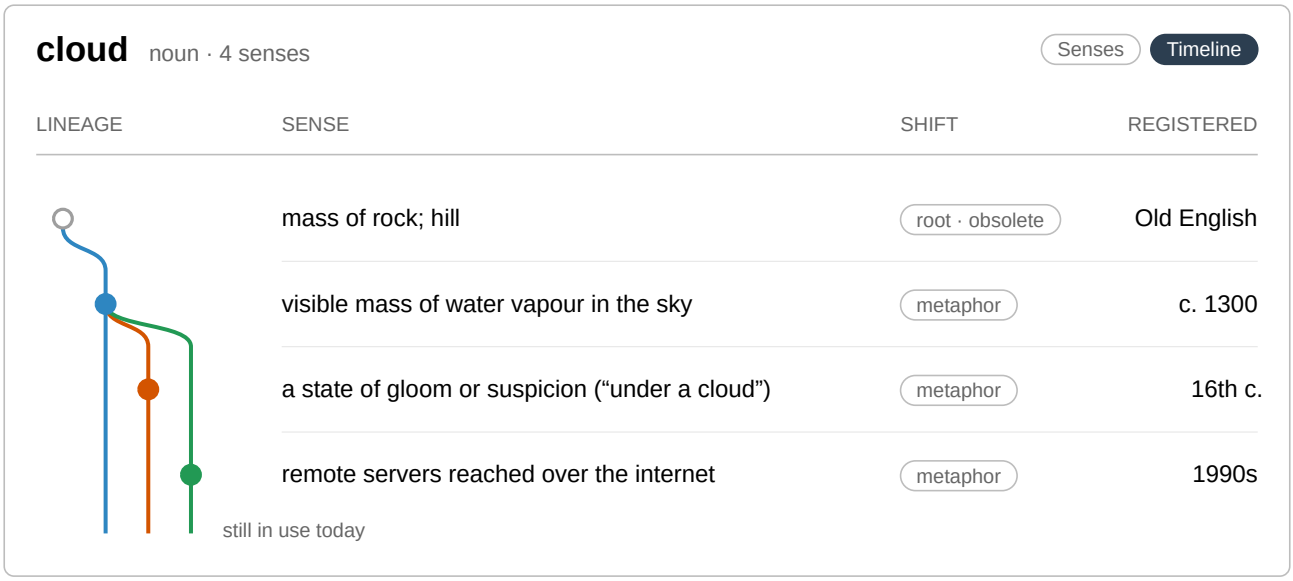


Figure 1: Wireframe. Each colour is one branch of meaning, a curve is a sense born from an earlier one, and a hollow node is an obsolete sense. Sample data for “cloud”; dates are illustrative.

5. Constraints

💡 *The non-negotiable rules the app must follow, on the interface and user-experience side.*

Constraint	Requirement
Desktop-first	Designed first for desktop browsers (research use), then adapted to tablet and phone
Performance	A word entry with up to 100 senses loads in under 2 seconds
Familiarity	Layout and terms follow en-word.net: synset, lemma, gloss, relation names

Constraint	Requirement
Graph readability	At most 8 branches side by side; extra branches fold into a “+n branches” marker
Accessibility	WCAG 2.1 AA; branches never differ by colour alone; every row states its parent sense in text; full keyboard navigation
Fuzzy dates	Dates may be exact, a year, a decade or a century (“c. 1300”, “16th c.”); rows sort by the earliest possible date and show the date as recorded
Provenance	Every date and parent link shows its source (dictionary, corpus, citation)
Open access	Browsing needs no account
Responsive	On phones the graph shrinks to a narrow strip and rows stay readable
Feedback	The user always sees what is happening (loading, empty result, error)
Language	Interface and lexicon in English

6. Out of Scope (MVP)

- No editing or submitting senses from the interface; the ShiftNet team loads the data
- No user accounts, favourites or saved searches
- No languages other than English
- No side-by-side comparison of two words’ timelines (planned for v2)
- No per-sense frequency charts from corpora
- No automatic detection of new senses in the interface
- No public API or bulk download from the interface

💡 ***Tip:*** *This PRD covers the WHAT (screens, features). The technical HOW (ShiftNet data format, database, APIs) comes next, once the interface is validated.*